The impact of e-banking service quality on e-customer satisfaction: Evidence from the undergraduates of the University of Ruhuna.

A proposal submitted to the Department of Mathematics of University of Ruhuna in partial fulfillment of the requirements for the award of the Honours Degree of Bachelor of Science in Financial Mathematics and Industrial Statistics

by
H.P.D. Poornima (SC/2022/12951)
B.M.S. Sachini Chamodya (SC/2022/12952)
G.P.A. Sewwandi Pathirana (SC/2022/12964)

Supervisor:
Dr. E.K. Jayampathi

2025.01.24

Table of Content

Relationship between the impact of e-banking service quality on e-customer satisfaction of students of University of Ruhuna .....

Abstract .....

Introduction .....

Literature Review .....

Methodology .....

- Research paradigm .....
- Conceptual framework .....
- Hypothesis .....
- Research site .....
- Population .....
- Sample .....
- Sample frame .....
- Sampling technique .....
- Sample size .....
- Data .....
- Data collection methods .....
- Data collection tools .....
- Data analysis .....
- Data presentation .....

Expected outcomes .....

References .....

Abstract

The main objective of this paper is to measure the impact of e-banking service quality on e-customer satisfaction based on undergraduates at the University of Ruhuna. Primary data method is used to collect data by using the questionnaire survey over 100 respondents who were randomly selected from the FMIS second year students. The Multiple regression technique has been employed to measure the effect of the E-Banking service quality dimension on E- Customer satisfaction. The study focuses on key dimensions of service quality such as Reliability, Responsiveness, Security and privacy, Web design and functionality.

Introduction

E-banking service quality is a key factor in customer satisfaction and the success of electronic banking services. Some dimensions of e-banking service quality such as reliability, responsiveness, security and privacy, web design and functionality are mainly affected to the customer satisfaction. Customer satisfaction is a measure of how happy customers are with a company's products, services, and capabilities. It's important for business success because satisfied customers are more likely to remain loyal, make repeat purchases, and refer others to the business.

High-quality e-banking services are essential for ensuring customer satisfaction. As technology advances, maintaining high service quality has become a essential thing for financial institutions in a competitive market. In the other hand, E-customer satisfaction is the degree to which customers are content with their overall experience when using e-banking services. Satisfied e-customers are mostly loyal to the bank, and recommend the platform to others.

As more customers turn to online banking for convenience and efficiency, the quality of these services becomes the primary factor in determining whether customers will continue to use the platform or switch to competitors. Because of that identifying this relationship is very useful.

Also, the relationship between e-banking service quality and e-customer satisfaction is significant for both academic, research, and practical applications. Especially the nature of this relationship remains unclear in certain demographic groups, particularly among university students. This study conducts to fill this gap by investigating the relationship between e-banking service quality and e-customer satisfaction among undergraduates at the University of Ruhuna.

For this purpose, we will conduct a survey and analyze main dimensions of service quality. And we believe, the findings of this study will offer valuable insights for banks in tailoring their e-banking services to better meet the needs of undergraduate students.

RESEARCH QUESTIONS

1. What is the impact of e-banking service quality on e-customer satisfaction?
2. What is the impact of reliability on e-customer satisfaction?
3. What is the impact of responsiveness on e-customer satisfaction?
4. What is the impact web design on e-customer satisfaction?
5. What is the impact of functionality on e-customer satisfaction?
6. What is the impact of privacy and security on e-customer satisfaction?

OBJECTIVE OF STUDY

1. To identify the impact of e-banking service quality on e-customer satisfaction.
2. To identify the impact of reliability on e-customer satisfaction.
3. To identify the impact of responsiveness on e-customer satisfaction.
4. To identify the impact web design on e-customer satisfaction.
5. To identify the impact of privacy and security on e-customer satisfaction.
6. To identify the impact of functionality on e-customer satisfaction.

Literature review

E-banking service quality refers to the overall perception of a bank's electronic services, which is influenced by various factors such as system performance, responsiveness, security, and user experience. Research by Parasuraman et al. (1988) and Zeithaml et al. (2002) identifies several dimensions of service quality, which are essential in determining customer satisfaction in e-banking platforms.

Reliability in e-banking refers to the consistency and dependability of the banking platform in performing transactions accurately and without interruption. According to Parasuraman et al. (1988), a reliable service ensures that users can complete their transactions smoothly and without errors, fostering trust and satisfaction. For university students, who often rely on e-banking for time-sensitive tasks like fee payments and fund transfers, reliability is crucial. Research by Alalwan et al. (2017) highlights that when students experience a reliable system, their confidence in using the service increases, leading to higher customer satisfaction and loyalty.

Responsiveness in e-banking refers to the speed and effectiveness with which a bank addresses customer inquiries, issues, and requests for assistance. Research by Zeithaml et al. (2002) shows that quick and efficient responses are key to customer satisfaction, especially for university students who rely on timely support for financial tasks. Alalwan et al. (2017) found that prompt customer service—whether through live chat, phone, or email—enhances the user experience by reducing frustration. Moreover, fast resolution of issues like transaction errors is crucial for building trust and loyalty (Sweeney et al., 1999).

Web design in e-banking refers to the user interface and visual appeal of the banking platform, which significantly affects user satisfaction. Research by Alalwan et al. (2017) emphasizes that a user-friendly and aesthetically pleasing design improves navigation, making it easier for university students to perform tasks such as bill payments or fund transfers. A well-designed platform reduces cognitive load and enhances overall satisfaction by providing a seamless experience. Narteh (2015) also highlights that a visually appealing interface is particularly important for younger users, like university students, who expect modern and intuitive digital experiences.

Functionality in e-banking refers to the essential features and services the platform offers, such as online transfers, bill payments, and account balance management. According to Zhang and Prybutok (2005), a well-functioning platform that supports a variety of services enhances customer satisfaction, as users expect to perform tasks quickly and efficiently. Alalwan et al. (2017) also note that for university students, whose financial activities are time-sensitive, the availability of reliable and easy-to-use features is crucial to ensure a positive user experience and continued engagement with the e-banking platform.

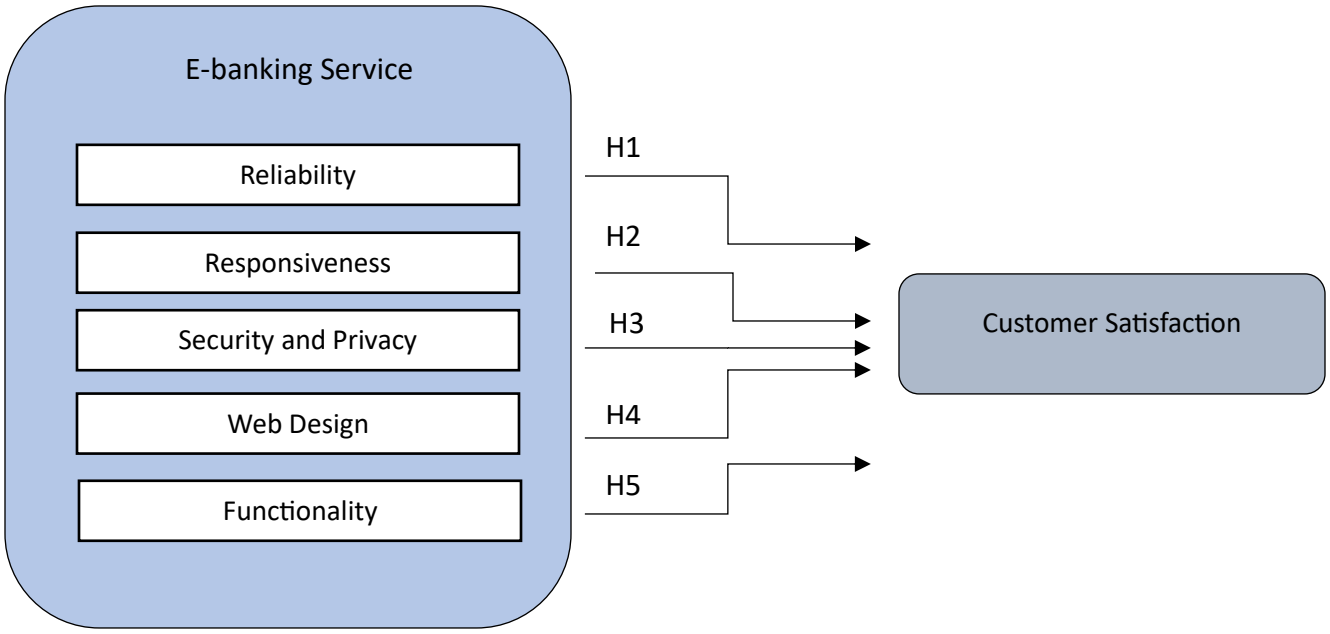
Security and **privacy** is critical dimensions of e-banking service quality, particularly for university students concerned about the safety of their personal and financial data. According to Lee et al. (2011), perceived security significantly impacts customer satisfaction, as users need assurance that their transactions are protected from fraud and unauthorized access. Furthermore, Chellappa and Sin (2005) argue that transparent privacy policies, which outline how personal data is collected and used, are essential for building trust.

RESEARCH METHODOLOGY

RESEARCH PARADIGM

This research follows a positivist paradigm to examine the impact of e-banking service quality on e-customer satisfaction among University of Ruhuna undergraduates. Using a quantitative approach, it focuses on measurable variables such as reliability, accessibility, and security. Structured questionnaires and statistical methods will test hypotheses and provide objective, generalizable insights into the relationship between service quality and customer satisfaction.

CONCEPTUAL FRAMEWORK



HYPOTHESIS

H1: Reliability positively influences e-customer satisfaction.

H2: Responsiveness positively influences e-customer satisfaction.

H3: Web design positively influences e-customer satisfaction.

H4: Privacy and security positively influences e-customer satisfaction.

H5: Functionality positively influences e-customer satisfaction.

The study adopts a quantitative research design to explore the relationship between e-banking service quality and customer satisfaction focused on the undergraduate students of Financial Mathematics and Industrial Statistics degree program in University of Ruhuna who actively use e-banking services.

Primary data method is used to collect data by using the questionnaire survey over a sample of 50 students from the FMIS second year students to ensure reliability and statistical validity. Secondary data will be drawn from the relevant research papers on the impact of e-banking service quality on customer satisfaction. The research will be conducted at University of Ruhuna. We plan to use online via Google Form / questioner to collect data and it will be analyzed using **SPSS** software.

The data is analyzed using descriptive statistics, correlation analysis, and regression analysis to explore relationships and impacts and the findings are presented using graphs, tables, and charts to illustrate trends and relationships clearly.

EXPECTED OUTCOMES

This study is expected to provide an understanding of how e-banking service quality influences e-customer satisfaction, with a specific focus on undergraduates at the University of Ruhuna. By analyzing key dimensions of service quality, the research aims to uncover the factors that contribute most significantly to customer satisfaction in the context of e-banking.

Previous studies have established a positive relationship between e-banking service quality and e-customer satisfaction. For instance, research by Parasuraman et al. (1988) highlights the role of service quality dimensions in enhancing customer satisfaction across industries, including banking. Similarly, a study by Al-Hawari et al. (2008) demonstrated that improved service quality in e-banking platforms significantly enhances customer satisfaction and loyalty. According to these findings, a study by Khan et al. (2021) on university students also indicated that responsiveness and reliability were key determinants of e-customer satisfaction in e-banking.

Drawing from these examples, we also expect a positive relationship between e-banking service quality and e-customer satisfaction in our study. The findings offer valuable insights for financial institutions seeking to enhance their e-banking services. Specifically, the study will help banks tailor their platforms to better meet the needs of young, tech-savvy users, thereby fostering greater customer satisfaction and loyalty.

Furthermore, the research will contribute to academic knowledge by addressing a gap in the literature regarding the relationship between e-banking service quality and e-customer satisfaction within a university demographic. This will also ensure a more user-centric approach in the highly competitive e-banking landscape.

REFERENCES

Sathiyavany N, Shivany S (2018), E-Banking Service Qualities, E-Customer Satisfaction, and e-Loyalty: A conceptual Model, The International Journal of Social Sciences and Humanities Invention, vol. 5, Issue 06, Jun, 2018

Abbas Keramati, Alireza Nili (2011), A Proposal Framework for investigating the impact of customer relationship management on customer retention in ecommerce, Proceedings of the 2011 International Conference on Industrial Engineering and Operations Management, Kuala Lumpur, Malaysia, January 22 – 24, 2011

Ali Naghi Mansour Beigi1, Saeid Jorfi, Reza Salemi Tajarrod, Hamed Mansour Beigi (2016), The Study of the Effects of E-banking Services Quality on Customers' Satisfaction and Loyalty (Case Study: Agricultural Bank of Khuzestan Province), journal of current research in science issn 2322-5009 coden (usa):

Adilah Binti Othman (2014), factors influencing the adoption of mobile banking among generation y
Samuel Godadaw Ayinaddis, Birhan Ambachew Taye,Bantie Gentnet Yirsaw (2023), Examining the effect of electronic banking service quality on customer satisfaction and loyalty: an implication for technological innovation.

nuradhi k. jayasiri, w.a.s.p. weerathunga (2008), popularity of e-banking in sri lanka
Jyotsna Sharma, Jagdeep Singh, Amandeep Singh (2020), Impact of E-Banking Service Quality on Customer Satisfaction, International Journal of Recent Technology and Engineering (IJRTE) ISSN: 2277 3878, Volume-8 Issue-5, January 2020.

Dr.K.Sathyabama, R.Samundeswari (2019), a study on customer satisfaction towards internet banking services in thoothukudi city.

Nur Lelasari, Innocentius Bernarto (2023-02-01), The Impact of Customer Satisfaction on Customer Loyalty of BCA Mobile Banking.

Shahid Kalim Khan, Najam-Ul-Hassan, Muhammad Naeem Anjum (2022), factors influencing customer satisfaction in e-banking services in Pakistan: evidence based on reliability, privacy & convenience
Liss Jenneboer , Carolina Herrando (2022), The Impact of Chatbots on Customer Loyalty: A Systematic Literature Review.

Jamil Hammoud, Rima M. Bizri, and Ibrahim El Baba (2018), The Impact of E-Banking Service Quality on Customer Satisfaction: Evidence from the Lebanese Banking Sector.

Do Thanh, Van Thanh, Dung Manh, Duyen Bich T (2020), Impact of Service Quality, Customer Satisfaction and Switching Costs on Customer Loyalty, Journal of Asian Finance, Economics and Business Vol 7 No 8 (2020) 395–405.

A.P.P. Perera, H.M.S. Priyanath (2018), The impact of internet banking service quality on customer Satisfaction, International Journal of Management, IT & Engineering, Vol. 8 Issue 2, February 2018.